Informatica® Metadata Command Center
October 2026

Snowflake SQL Script Sources

Informatica Metadata Command Center Snowflake SQL Script Sources
October 2026

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Table of Contents

Preface 4

Chapter 1: Introduction to Snowflake SQL Script catalog sources 5

Extraction and view process 6

About the Snowflake SQL Script catalog source 6

Extracted metadata 6

Compatible connectors 7

Chapter 2: Before you begin 8

Verify permissions 8

Permissions for metadata extraction 8

Create a connection 9

Standard authentication 10

Authorization code authentication 11

Key pair authentication 14

Client credentials 15

Programmatic access token authentication 16

Chapter 3: Create catalog sources in Metadata Command Center 18

Step 1. Register a catalog source 18

Step 2. Configure capabilities 20

Configure metadata extraction 20

Configure automation 22

Step 3. Associate stakeholders and asset groups 24

Step 4. Run or schedule the job 26

Step 5. Assign reference catalog source connections to endpoint catalog source objects 27

Chapter 4: View results in Data Governance and Catalog 29

View metadata extraction results 30

View data lineage 31

View lineage at the catalog source level 32

View lineage at the data set level 32

View lineage at the data element level 32

Appendix A: Managing catalog sources 34

Cloning a catalog source configuration 34

Deleting or purging a catalog source 35

Purging obsolete objects from a catalog source 35

Curating CLAIRE recommendations 36

Managing configuration permissions for catalog sources 36

Preface

Read *Snowflake SQL Script Sources* to learn how to register and configure Snowflake SQL Script sources as catalog sources in Metadata Command Center. After you configure a catalog source, you extract metadata and then view the results in Data Governance and Catalog.

CHAPTER 1

Introduction to Snowflake SQL Script catalog sources

You can use Metadata Command Center to extract metadata from a source system.

A source system is any system that contains data or metadata. For example, Snowflake database is a source system from which you can extract metadata through a Snowflake SQL Script catalog source with Metadata Command Center. A catalog source is an object that represents and contains metadata from the source system.

Before you extract metadata from a source system, you first create and register a catalog source that represents the source system. Then you configure capabilities for the catalog source. A capability is a task that Metadata Command Center can perform, such as metadata extraction, lineage discovery, autonomous enrichment, data profiling, data classification, or glossary association.

When Metadata Command Center extracts metadata, Data Governance and Catalog displays the extracted metadata and its attributes as technical assets. You can then perform tasks such as analyzing the assets, viewing lineage, and creating links between those assets and their business context.

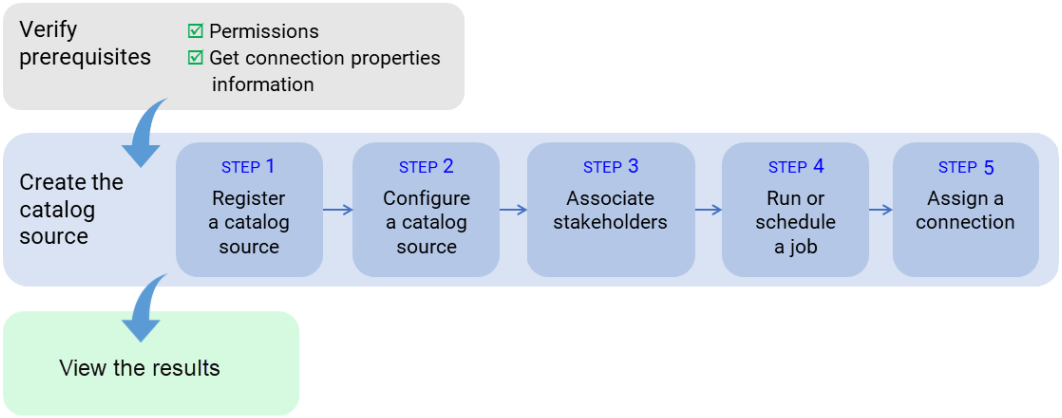
The following table describes the capabilities of the catalog source:

Capability	Description
Lineage Discovery	Builds the complete lineage of a catalog source by recommending endpoint catalog source objects to assign to reference catalog source connections. When you run the catalog source job, Metadata Command Center assigns the reference catalog source connections to CLAIRE recommended endpoint catalog source objects. You can then view the list of CLAIRE recommendations and accept or reject them.
Autonomous Enrichment	Automatically propagates classification, description, business name, and glossary enrichments from an asset to related assets based on the lineage between assets. Autonomous enrichment reduces manual curation effort by extending the enrichments that you apply to one asset across assets that are connected to it through lineage.

Extraction and view process

You can use a Snowflake SQL Script catalog source to extract metadata from script files that define transformations on an Oracle Database. Configure the catalog source and run the extraction job in Metadata Command Center. Then view the results in Data Governance and Catalog.

The following image shows the process to extract metadata from a script:



After you verify prerequisites, perform the following tasks to extract metadata from Snowflake SQL Script:

1. Register a catalog source. Create a catalog source object, select Snowflake SQL Script, and specify values for connection properties.
2. Configure the catalog source. Specify the runtime environment and add filters.
3. Associate stakeholders. Optionally, associate users with technical assets, giving the users permission to perform actions determined by their roles.
4. Run or schedule the catalog source job.
5. Assign a connection to referenced source system assets.

About the Snowflake SQL Script catalog source

You can use the Snowflake SQL Script catalog source to extract metadata from the Snowflake source system.

Snowflake SQL Script is a set of Snowflake SQL statements stored in files that you can use for sequential running of scripts.

Extracted metadata

You can extract specific metadata from Snowflake SQL scripts with a Snowflake SQL Script catalog source.

Metadata Command Center extracts the following metadata from the Snowflake SQL Script source system:

- Calculation
- Folder

- Script
- Statements

Compatible connectors

Before you configure Snowflake SQL Script catalog source, you must connect to the source system.

Use the Snowflake Data Cloud connector to connect to the Snowflake source system. For information about configuring a connection, see *Connections* in the Administrator help system.

CHAPTER 2

Before you begin

Before you can extract catalog source metadata, get information from the Snowflake database administrator.

Ensure that the following prerequisites are met:

- You have Snowflake SQL Script files saved on the runtime environment from which you want to extract metadata.
- You have access to Snowflake SQL Script.
- You have database source information to connect to the Snowflake database.
- You use the Snowflake Data Cloud connector.
- Verify permissions.

Verify permissions

To extract metadata and to configure other capabilities that a catalog source might include, you need account access and permissions on the source system. The permissions required might vary depending on the capability.

Permissions for metadata extraction

Ensure that you have the required permissions to enable metadata extraction.

Verify that you have the following permissions:

- READ and EXECUTE permissions on folders to access and traverse the folders
- READ permission on files
- SELECT permissions on system tables for the Snowflake database user account:

    select on information_schema.EXTERNAL_TABLES

    select on information_schema.FUNCTIONS

    select on information_schema.PIPES

    select on information_schema.PROCEDURES

    select on information_schema.SCHEMATA

select on information_schema.SEQUENCES

select on information_schema.STAGES

select on SNOWFLAKE.ACCOUNT_USAGE.TAGS

select on SNOWFLAKE.ACCOUNT_USAGE.TAG_REFERENCES

- SHOW permissions on database objects for the Snowflake database user account:

  show columns

  show primary keys

  show imported keys

  show streams

  show materialized views

  show tasks

  show databases

  show objects

Create a connection

When you configure a connection, you specify the connection properties for the connection. Connection properties enable an agent to connect to data sources.

1. In Administrator, select **Connections**.
2. Click **Add Connection**.
3. Select Snowflake Data Cloud and click **Next**.
4. Enter the following connection details:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	Snowflake Data Cloud

5. Select the runtime environment where you want to run the tasks. The runtime environment is either a Secure Agent or a serverless runtime environment.
6. Select the authentication method.

   You can use the following authentication methods to connect to Snowflake:

   - Standard. Uses the Snowflake account user name and password credentials to connect to Snowflake.
   - Authorization Code. Uses the Snowflake account name, warehouse name, authorization URL, access token URL, client ID, client secret, and access token to connect to Snowflake.

- Key Pair. Uses the private key file and private key file password, along with the existing Snowflake account user name to connect to Snowflake.
- Client Credentials. Uses the client ID, access token URL, client secret, scope, and the access token at a minimum.
- Programmatic Access Token. Uses the user name, account name, warehouse name, and programmatic access token to connect to Snowflake.

7. Click **Test** to test your connection to the source system.
8. Click **Save**.

Standard authentication

This authentication method requires the Snowflake account user name and password credentials to connect to Snowflake.

The following table describes the basic connection properties for standard authentication:

Property	Description
Username	The user name to connect to the Snowflake account.
Password/PAT	The password or programmatic access token (PAT) to connect to the Snowflake account. Enter the password of your Snowflake account or the PAT granted by Snowflake. For more information about how to generate a PAT, see *Generate the programmatic access token*.

Property	Description
Account	The name of the Snowflake account. For example, if the Snowflake URL is `https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/`, your account name is the first segment in the URL before snowflakecomputing.com. Here, `123abc.us-east-2.aws` is your account name. If you use the Snowsight URL, for example, `https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard`, your account name is `123abc.us-east-2.aws`. **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.

Property	Description
Role	The Snowflake role assigned to the user.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	You can specify multiple JDBC connection parameters, separated by ampersand (&), in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, you can pass the following database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	When you add parameters, ensure that there is no space before and after the equal sign (=).

Authorization code authentication

OAuth 2.0 authentication requires the OAuth 2.0 protocol with Authorization Code grant type to connect to Snowflake. Authorization Code allows authorized access to Snowflake without the need to share or store your login credentials.

The following table describes the basic connection properties for OAuth 2.0 authorization code authentication:

Property	Description
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.
Authorization URL	The Snowflake server endpoint that is used to authorize the user request. The authorization URL is https://<account name>.snowflakecomputing.com/oauth/authorize, where <account name> specifies the full name of your account provided by Snowflake. For example, https://<abc>.snowflakecomputing.com/oauth/authorize **Note:** If the account name contains underscores, use the alias name. You can also use the Authorization Code grant type that supports the authorization server in a Virtual Private Cloud network.
Access Token URL	The Snowflake access token endpoint that is used to exchange the authorization code to get an access token. The access token URL is https://<account name>.snowflakecomputing.com/oauth/token-request, where <account name> specifies the full name of your account provided by Snowflake. For example, https://<abc>.snowflakecomputing.com/oauth/token-request **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.
Client ID	Client ID of your application generated when you create a security integration of type OAuth in Snowflake.

Property	Description
Client Secret	Client secret generated for the client ID.
Access Token	The access token value. Enter the populated access token value that you get from the OAuth endpoint, or click **Generate Access Token** to populate the access token value.

Advanced settings

The following table describes the advanced connection properties for OAuth 2.0 authorization code authentication:

Property	Description

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	You can specify multiple JDBC connection parameters, separated by ampersand (&), in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, you can pass the following database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	When you add parameters, ensure that there is no space before and after the equal sign (=).

Scope	Determines the access control when the API endpoint has defined custom scopes. For example, specify `session:role:CQA_GCP` as the scope to override the value of the default user role. The value needs to be one of the roles assigned in Security Integration. To enter multiple scope attributes, separate each scope attribute with a space.

Access Token Parameters	Additional parameters to use with the access token URL.
	Define the access token parameters in the following JSON format:
	`[{"Name":"<Parameter name>","Value":"<Parameter value>"}]`
	For example, you can use the following code_verifier parameter when you connect to Snowflake:
	`[{"Name":"code_verifier","Value":"5PMddu6Zcg6Tc4sbg"}]`
	For more information about access token parameters that you can define, see *Introduction to OAuth* in the Snowflake documentation.

Property	Description

Authorization Code Parameters	Additional parameters to use with the authorization token URL.
	Define multiple parameters, separated by comma, in the following JSON format:
	`[{"Name":"<Parameter name>","Value":"<Parameter value>"}, {"Name":"<Parameter name>","Value":"<Parameter value>"}]`
	For example, you can use the following code_challenge and code_challenge_method parameters when you connect to Snowflake:
	`[{"Name":"code_challenge","Value":"Ikr-vv52th0UeVRi4"}, {"Name":"code_challenge_method","Value":"S256"}]`

Refresh Token	The refresh token value. Enter the populated refresh token value that you get from the OAuth endpoint, or click **Generate AccessToken** to populate the refresh token value. If the access token is not valid or expires, the Secure Agent fetches a new access token with the help of the refresh token. **Note:** If the refresh token expires, provide a valid refresh token or regenerate a new refresh token by clicking **Generate Access Token**.

Key pair authentication

This authentication method requires the private key file and private key file password, along with the existing Snowflake account user name to connect to Snowflake.

The following table describes the basic connection properties for key pair authentication:

Property	Description
Username	The user name to connect to the Snowflake account.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.

Property	Description
Warehouse	The Snowflake warehouse name.
Private Key File	Path to the private key file, including the private key file name, that the Secure Agent uses to access Snowflake. For example, specify the following path and key file name in the Secure Agent machine: - On Windows: `C:\Users\path_to_key_file\rsa_key.p8` - On Linux: `/export/home/user/path_to_key_file/rsa_key.p8` To use the serverless runtime environment, specify the following path and key file name in the serverless agent directory: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<Private key file name>` **Note:** Verify that the keystore is FIPS-certified.

The following table describes the advanced connection properties for key pair authentication:

Property	Description

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	You can specify multiple JDBC connection parameters, separated by ampersand (&), in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, you can pass the following database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	When you add parameters, ensure that there is no space before and after the equal sign (=).

Private Key File Password	This property is required when you enable the secret vault in the connection. Password for the private key file.

Client credentials

The OAuth 2.0 client credentials authentication requires at a minimum the client ID, access token URL, client secret, scope, and the access token.

The following table describes the basic connection properties for OAuth 2.0 client credentials authentication:

Property	Description
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.

Property	Description
Access Token URL	The Snowflake access token endpoint that is used to exchange the authorization code for an access token. Specify the access token URL that you get from the OAuth endpoint.
Client ID	Client ID of your application generated when you configure the application for OAuth.
Client Secret	Client secret generated for the client ID.
Scope	Determines the access control when the API endpoint has defined custom scopes. For example, specify `session:role:CQA_GCP` as the scope to override the value of the default user role. The value needs to one of the roles assigned in Security Integration. To enter multiple scope attributes, separate each scope attribute with a space.
Access Token	The access token value. Enter the populated access token value that you get from the OAuth endpoint, or click **Generate Access Token** to populate the access token value.

The following table describes the advanced connection properties for the OAuth 2.0 client credentials authentication:

Property	Description

Additional JDBC URL Parameters	Additional JDBC connection parameters.
	You can specify multiple JDBC connection parameters, separated by ampersand (&), in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, you can pass the following database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	When you add parameters, ensure that there is no space before and after the equal sign (=).
	For the list of additional JDBC parameters that you can configure, see "JDBC URL parameters" in *Snowflake Data Cloud connection properties* in the Administrator help.

Access Token Parameters	Additional parameters to use with the access token URL.
	Define the access token parameters in the following JSON format:
	`[{"Name":"<Parameter name>","Value":"<Parameter value>"}]`
	For example, you can use the following code_verifier parameter when you connect to Snowflake:
	`[{"Name":"code_verifier","Value":"5PMddu6Zcg6Tc4sbg"}]`
	For more information about access token parameters that you can define, see *Introduction to OAuth* in the Snowflake documentation.

Programmatic access token authentication

The minimum requirement for an programmatic access token authentication is the Snowflake user name, programmatic access token, account name, and warehouse name.

The following table describes the basic connection properties for programmatic access token authentication:

Property	Description
Username	The user name to connect to the Snowflake account.
Programmatic Access Token	The programmatic access token to connect to the Snowflake account. Enter the programmatic access token granted by Snowflake.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.

Advanced settings

The following table describes the advanced connection properties for programmatic access token authentication:

Property	Description
Role	The Snowflake role assigned to the user.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	You can specify multiple JDBC connection parameters, separated by ampersand (&), in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, you can pass the following database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	When you add parameters, ensure that there is no space before and after the equal sign (=).
	For the list of additional JDBC parameters that you can configure, see "JDBC URL parameters" in *Snowflake Data Cloud connection properties* in the Administrator help.

CHAPTER 3

Create catalog sources in Metadata Command Center

Use Metadata Command Center to configure a catalog source for Snowflake SQL Script and run the metadata extraction job.

When you configure a catalog source, you define the source system from which you want to extract metadata. Configure filters to include or exclude source system metadata before you run the job.

To provide stakeholders access to technical assets, you can assign access through roles. To view lineage for any system that the source system references, create a catalog source and a connection associated with the referenced source system after you run the job.

Step 1. Register a catalog source

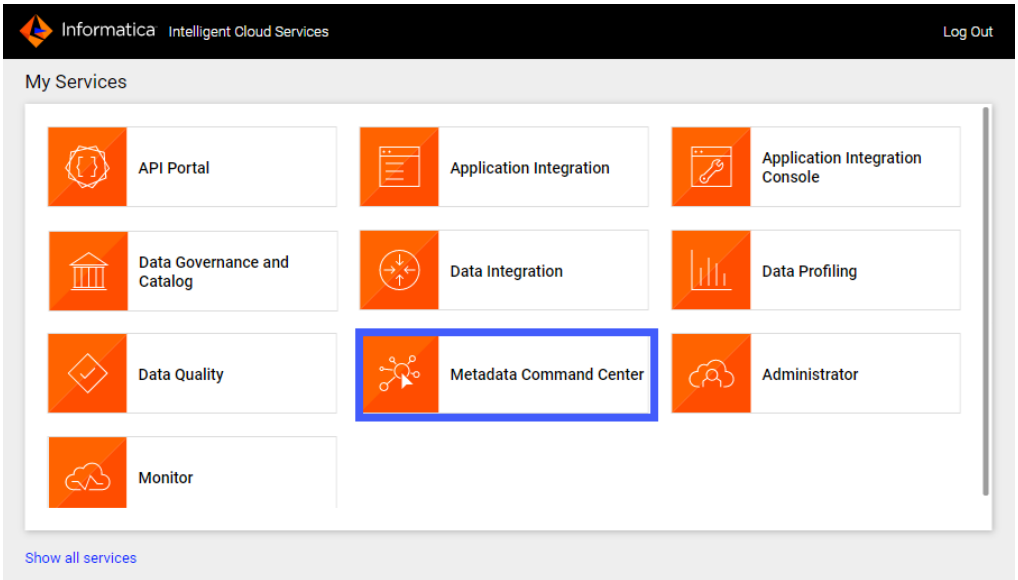
When you register a catalog source, provide general information and connection values.

1. Log in to Intelligent Data Management Cloud.

   The **My Services** page appears.

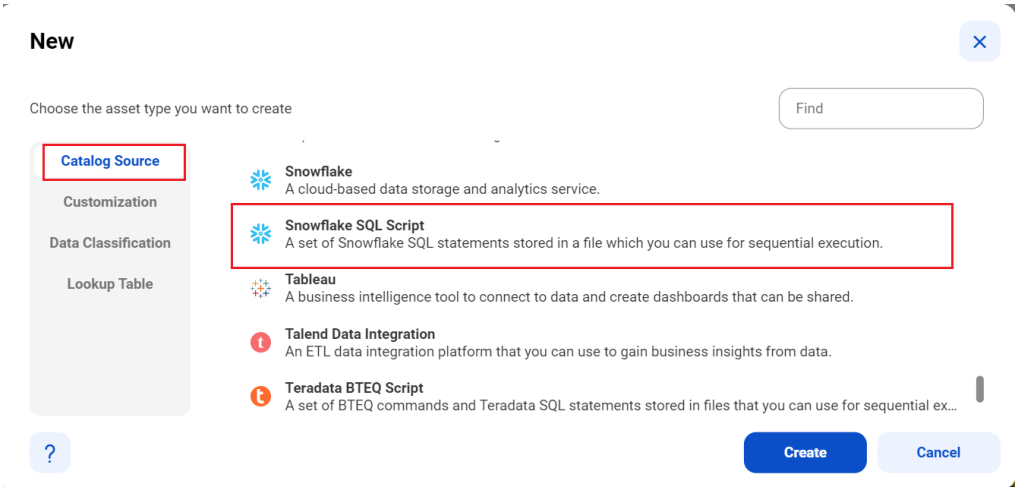
2. Click **Metadata Command Center**.

   The **Metadata Command Center** box appears on the **My Services** page.



The Metadata Command Center home page appears.

3. Click **New**.
4. Select **Catalog Source** from the list of asset types.
5. Select Snowflake SQL Script from the list of source systems.



6. Click **Create**.

   The **New Catalog Source** page opens.

7. In the **General Information** section, enter a name and an optional description for the catalog source.

   Note: You can rename a catalog source after you create it, but to apply the change to all associated objects you must rerun the metadata extraction job.

   After you save the catalog source, you can update the description in Metadata Command Center and Data Governance and Catalog. The update appears only in the service in which you update it.

8. In the **Connection Information** area, select the connection that you created in Administrator.

   Note: To create or edit a catalog source, you need permissions on the connection to the source system. Select a connection that you have access to, or ask the administrator to grant the necessary permissions to the connection that you want to use.

9. Click **Connection Properties** to expand and view the connection properties for the selected connection.
10. Click **Test Connection** to test your connection to the source system.
11. Click **Next**.

The **Configuration** page appears.

Step 2. Configure capabilities

When you configure the Snowflake SQL Script catalog source, you define the settings for the metadata extraction capability.

The metadata extraction capability extracts source metadata from external source systems. You can also configure other capabilities that the catalog source includes.

You can save the catalog source configuration at any point after you enter the connection information. After you save the catalog source, you can choose to run the catalog source job. To run the job once, click **Run**. To run metadata extraction and other capabilities on a recurring schedule, configure schedules on the **Schedule** tab.

You can save the catalog source configuration at any point after you enter the connection information. After you save the catalog source, you can choose to run the catalog source job. To run the job once, click **Run**. To run metadata extraction and other capabilities on a recurring schedule, configure schedules on the **Schedule** tab.

Configure metadata extraction

When you configure the Snowflake SQL Script catalog source, you choose a runtime environment and enter configuration parameters for metadata extraction.

1. In the **Connection and Runtime** area, choose a serverless runtime environment or the Secure Agent group where you want to run catalog source jobs.

   Note: Serverless runtime environment options are available if the catalog source works with a serverless runtime environment.
2. Choose to retain, delete, or deprecate objects that are deleted from the source system in the catalog with the **Metadata Change Option**.
   - **Retain**. Retains objects that are deleted from the source system in the catalog. If you update or add a filter, the catalog retains objects extracted from the previous job and extracts additional objects that match the current filter. Objects deleted from the source system are not deleted from the catalog. Enrichments added on deleted objects and relationships are retained.
   - **Delete**. Deletes metadata from the catalog based on objects deleted from the source system and changes you make to the filter. Enrichments added on deleted objects and relationships are also permanently lost. Objects renamed in the source system are removed and recreated in the catalog.

- **Deprecate**. When objects are deleted from the source system, the catalog changes the lifecycle of the imported objects to Obsolete. The lifecycle of objects also changes to Obsolete when you add exclusion filters or reduce the scope so that objects extracted in earlier jobs are no longer included. For example, the first job includes the sales schema. The second job also includes the sales schema, but excludes the sales.transactions table. In this case, the second job marks the sales.transactions object as Obsolete, even though the first job includes it. This does not impact enrichments applied to deprecated objects and relationships. Objects renamed in the source system are removed and recreated in the catalog. When you run the catalog source job again for other capabilities such as data classification, relationship discovery, or glossary association, the job doesn't consider obsolete objects. Obsolete objects remain in the catalog until they are purged when you run a **Purge Obsolete Objects** job on the **Explore** page.

Note: You can also change the configured metadata change option when you run a catalog source.

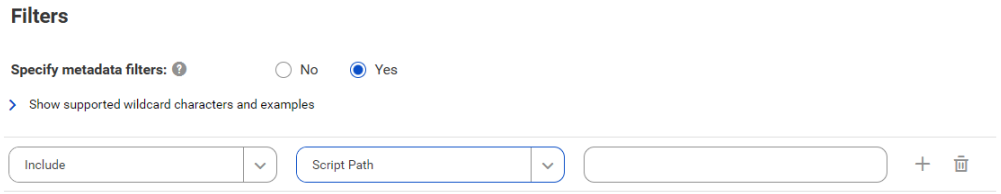
3. In the **Filters** area, define one or more filter conditions to apply to metadata extraction.
   a. Select **Yes** to view filter options.
   b. From the Include/Exclude list, choose to include or exclude metadata based on the filter parameters.
   c. From the **Select the object type** list, select **Script Path**.
   d. In the **Enter a value to specify the object location** dialog box, enter the script path as the filter value.

   Filter values can contain the following wildcard characters:

   - An asterisk to represent multiple characters.
   - A question mark to represent a single character.
   - If an object contains an asterisk or a question mark, enclose the symbol in double quotes.
   - If a filter value contains spaces before or after the string value, enclose the value in double quotes.

   Don't use wildcard characters in file paths. To enter a path hierarchy, use separators, such as a period or a slash, which the source system allows.

   The following image shows the **Filter conditions** options:



   If the scripts root directory path is `/users/opt/input`, use the following examples to create filter conditions:

   - To include or exclude metadata from the script named script1.sql located in the path `/users/opt/input/folder1/`, enter: `folder1/script1.sql`
   - To include or exclude metadata from all scripts with SQL extension stored in the path `/users/opt/input/folder1/`, enter: `folder1/*.sql`
   - To include or exclude metadata from all scripts stored in the path `/users/opt/input/`, enter: `*`
   - To include or exclude metadata from all scripts with SQL extension and names that start with 'script' followed by a single character, stored in the `path /users/opt/input/folder1/`, for example, `folder1/script1.sql, folder1/script2.sql, enter: folder1/script?.sql`
   - To include or exclude metadata from the script named script1.sql located in a directory with a name that starts with 'folder1' followed by zero or more characters, enter: `folder1*/script1.sql`

- To include or exclude metadata from all scripts with SQL extensions from folder2 as well as any subfolders within folder2, enter: `folder1/folder2/**/*.sql`
- To include or exclude metadata from all scripts from folder2 as well as any subfolders within folder2, enter: `folder1/folder2/*, folder1/folder2/, or folder1/folder2`

4. In the **Configuration Parameters** area, enter configuration parameters.

   The following table describes the properties that you can enter:

Property	Description
Scripts Root Directory Path	Path to the remote SQL script root directory.
Default Database	Default database for the SQL script processing.
Default Schema	Default schema for the SQL script processing.
MetaTables Include Filter	Advanced parameter. When you process PL/SQL statements, Metadata Command Center does not read tables or view content by default. If you want to use the content, for example, to process dynamic SQL statements, use the **MetaTables Include Filter** parameter. This parameter prompts the database for the required metadata. Verify that the user has SELECT permissions for metatables. **Note:** This parameter appears when you click **Show Advanced**. **Note:** Don't use this option to specify filters for tables that you want to include or exclude during the metadata extraction run.
Additional Settings	Configure expert parameters to specify additional configuration options to be passed at runtime. Required if you need to troubleshoot the catalog source job. **Caution:** Use expert parameters when it is recommended by Informatica Global Customer Support. **Note:** This parameter appears when you click **Show Advanced**.

5. Click **Next**.

   The **Associations** page appears.

Configure automation

Enable the lineage discovery and autonomous enrichment capabilities to build complete lineage using CLAIRE and to receive enrichments propagated from catalog sources connected through lineage.

Enable Lineage-based Propagation on the Autonomous Enrichment tab of the Configure page before you configure this catalog source to receive enrichments. For more information, see *Administration*.

With the lineage-based propagation settings on this tab, you configure how this catalog source receives enrichments propagated from connected catalog sources. Enrichments propagate only to assets that don't already have enrichments.

1. Click the **Automation** tab.
2. In the **Lineage Discovery** section, select **Enable Lineage Discovery**.
3. In the **Filters** area, define one or more filter conditions to apply for lineage discovery.

To define filters, you can choose to select catalog source types, asset groups, or enter a catalog source name or search from a list of catalog sources.

a. Select **Yes** to view filter options.

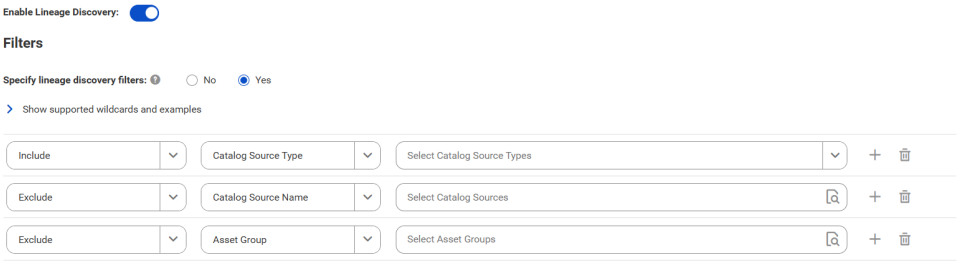
b. From the Include/Exclude list, choose to include or exclude catalog sources for lineage discovery based on the filter parameters.

c. From the filter type list, select catalog source type, catalog source name, or asset group.

d. In the filter value field, select the required catalog source types, or click the Search button and select catalog sources or asset groups.

Filters can contain the asterisk wildcard to represent multiple characters or empty text.

The filter options appear.



Examples:

- To include or exclude all Oracle catalog sources, select **Catalog Source Type** as the filter type and select `Oracle` in the filter value field.
- To include or exclude the 'Oracle_Retail' catalog source, select **Catalog Source Name** as the filter type and search for the catalog source or enter `Oracle_Retail` in the filter value field.
- To include or exclude all catalog sources with names that start with 'Oracle', select **Catalog Source Name** as the filter type and search for the catalog source or enter `Oracle*` in the filter value field.
- To include or exclude all catalog sources with names that end with 'Retail', select **Catalog Source Name** as the filter type and search for the catalog source or enter `*Retail` in the filter value field.
- To include or exclude all catalog sources with names that contain 'Ret', select **Catalog Source Name** as the filter type and search for the catalog source or enter `*Ret*` in the filter value field.
- To include or exclude all catalog sources that are part of the 'Financial Group' asset group, select **Asset Group** as the filter type and search `Financial Group` in the filter value field.

Note: You can't add more than one include or exclude filter for the same filter type.

e. Optionally, to define an additional filter with an AND condition, click the **Add** icon.

4. In the **Autonomous Enrichment** section, select **Enable Lineage-based Propagation**.

The lineage-based propagation settings appear.

5. Select the enrichment types to receive from connected catalog sources.

You can select Classification, Description, Business Name, and Glossary Term.

6. Choose how to apply received enrichments.

- Select **Auto-accepted** to apply received enrichments automatically.
- Select **Recommended** to list them as suggestions that you can curate in Data Governance and Catalog.

7. Choose whether to receive enrichments only from certified assets.

Assets are certified in Data Governance and Catalog. This option is selected by default. If you clear it, this catalog source also receives enrichments from assets that are not certified, which might be less reliable.

8. Choose where this catalog source receives enrichments from.
   - Select **Catalog Source** to receive enrichments from a specific catalog source.
   - Select **Nearest Neighbor** to receive enrichments from the closest connected catalog source based on lineage.
9. If you select **Catalog Source**, select a catalog source from the list.

   Ensure that the catalog source you select is within 15 lineage hops. Enrichments don't propagate from a catalog source beyond that distance.
10. If you select **Nearest Neighbor**, configure the lineage settings:
    a. Set **Lineage Direction** to search upstream, downstream, or in both directions.

       Upstream catalog sources feed data into this catalog source. Downstream catalog sources receive data from this catalog source. Select both directions to search upstream and downstream at the same time.

    b. Set **Number of Hops** to define how far along lineage to search.

       The search stops at the first connected catalog source that has enrichments, up to a maximum of 15 hops.
11. Click **Save**.

When you save, the catalog source begins receiving enrichments created from that point onward. Saving doesn't retrieve enrichments created prior to this date and time. To receive such enrichments, run the catalog source job manually and include the Autonomous Enrichment capability.

Step 3. Associate stakeholders and asset groups

Associate users or user groups within a stakeholder role as stakeholders for technical assets in Data Governance and Catalog. Also, you can choose to assign technical assets extracted from the catalog source to asset groups. You can then use access policies to control permissions on assets that are assigned to asset groups.

Verify that the administrator assigned users and user groups to the stakeholder role that you want to associate with technical assets.

1. To associate users or user groups as stakeholders with technical assets extracted from the catalog source, perform the following steps:
   a. On the **Associations** page, click **Stakeholders**.
   b. Select **Assign Stakeholders**.
   c. Select **Propagate to Child Assets** to propagate stakeholders of a parent asset to child assets without stakeholders.
   d. Select **Append Stakeholders** to propagate stakeholders of a parent asset to all child assets including assets with stakeholders assigned.

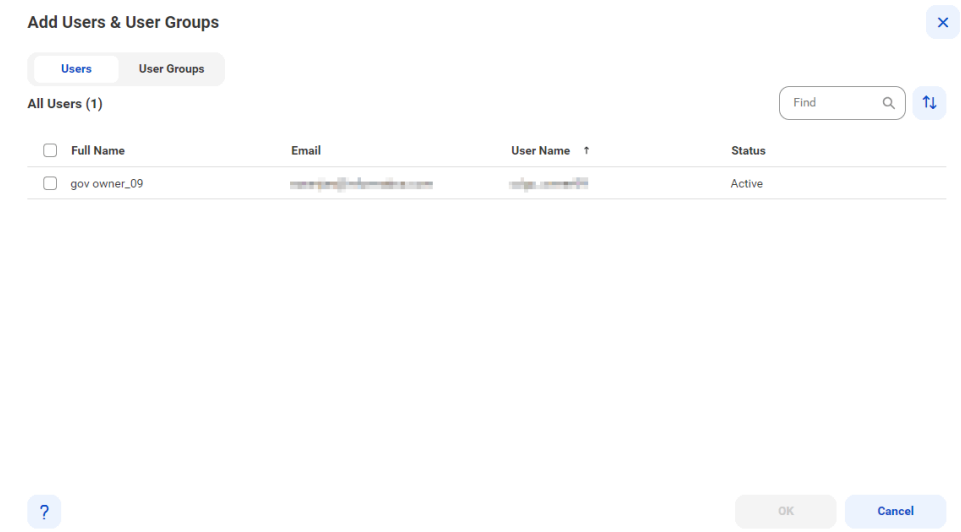
      Note: To enable **Append Stakeholders**, you must enable **Propagate to Child Assets**.

You can also override the propagation settings from the catalog source page in Data Governance and Catalog. For more information about stakeholder propagation, see *Stakeholder Propagation* in *Administration*.

e. Select a stakeholder role.

f. Click **Select** to add users and user groups from the stakeholder role as stakeholders for the technical assets.

The **Add Users & User Groups** dialog box displays a list of users and user groups assigned to the selected stakeholder role.



g. Select one or more users or user groups to assign as stakeholders for the technical assets, and click **OK**.

Only the selected users and user groups belonging to the specified stakeholder role are granted the permissions to technical assets.

h. To assign users or user groups from another stakeholder role, click **Add** and then repeat the steps.

2. To assign asset groups to technical assets extracted from the catalog source, perform the following steps:

a. On the **Associations** page, click **Asset Groups**.

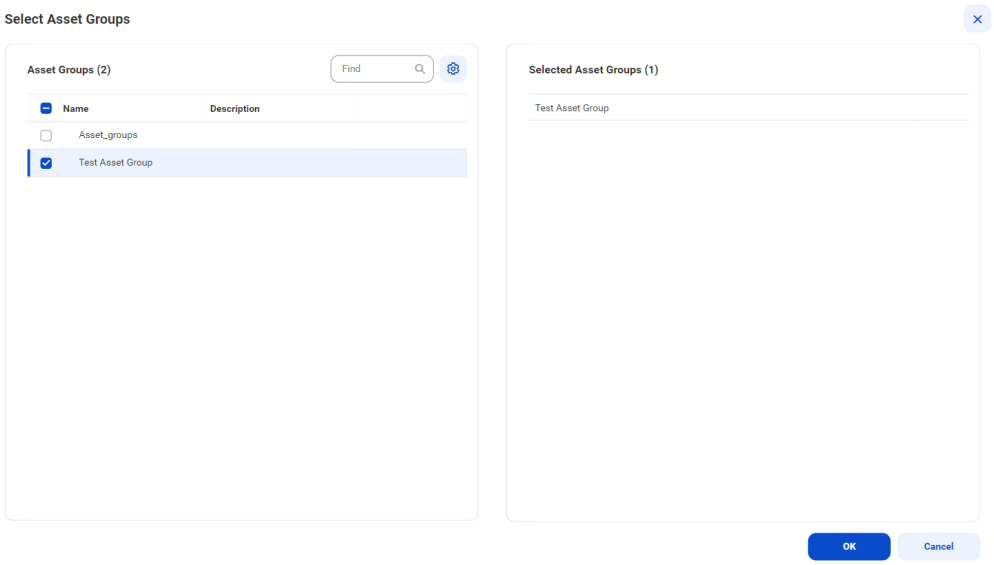
b. Select **Assign Asset Groups**.

c. Click **Select**.

The **Select Asset Groups** dialog box displays the list of asset groups.

If you enabled an access policy that includes an asset group, you can only view assets that belong to that asset group.

3. Select the asset groups to which you want to assign technical assets extracted from the catalog source, and click **OK**.



4. Choose to save and run the job or to schedule a recurring job.
   - To save and run the job, click **Save** and then **Run**.
   - To schedule a recurring job, click **Next** to open the **Schedule** page.

Step 4. Run or schedule the job

Choose to run a catalog source job manually, or configure it to run on schedule.

Note: You can't run multiple jobs simultaneously.

You can choose to perform a full or an incremental metadata extraction. A full metadata extraction extracts all objects from the source to the catalog. An incremental metadata extraction extracts only the changed and new objects since the last successful catalog source job run. Incremental metadata extraction doesn't remove deleted objects from the catalog and doesn't extract metadata of code-based objects if applicable.

When you run an incremental metadata extraction job with a filter to include metadata from objects, the job extracts only the objects that have the latest timestamp since the last successful job.

Note: If incremental metadata extraction is available for the catalog source, the option is always visible, but incremental metadata extraction jobs can run only after a scheduled full metadata extraction job. Manual jobs do not enable the incremental metadata extraction option.

Run the job manually

Click **Save** to save the catalog source and click **Run**. On the **Run Catalog Source Job** window, click **Run** to run the job.

You can override the capabilities that you selected while configuring your catalog source on the **Configuration** page. The first time you run the catalog source job, the metadata extraction capability is mandatory. From the second run onwards, you can choose to override the configured metadata change option. You can retain, delete, or deprecate objects that are deleted from the source in the catalog. For subsequent runs of the catalog source job, the metadata extraction capability is optional.

Note: You can choose incremental metadata extraction for subsequent runs only after one full metadata extraction job completes successfully. Incremental metadata extraction jobs run with the **Retain** metadata change option even if you set the option to **Delete** or **Deprecate** in the catalog source.

Note: To run a catalog source job, you need permissions on the connection to the source system. To run a catalog source job for catalog sources that reference other source systems, you need permissions on the connections for all the reference source systems.

Run the job on a schedule

You can choose to run metadata extraction and other capabilities on a recurring schedule. You can't choose incremental metadata extraction and full metadata extraction in the same schedule. To create a schedule for incremental metadata extraction, you must have completed at least one full metadata extraction job successfully. If not, first create a schedule for a full metadata extraction.

If an incremental metadata extraction is scheduled to run when the last run details aren't available, the job first performs a full metadata extraction, followed by incremental metadata extraction on subsequent runs.

For example, this can happen in the following scenarios:

- You create schedules for both incremental metadata extraction and full metadata extraction, but schedule the incremental extraction to run before the first full metadata extraction job.
- You create schedules for both incremental metadata extraction and full metadata extraction, but delete the full metadata extraction schedule before its first run.

1. On the **Schedule** tab, select **Run on Schedule**.
   The **Schedule** configuration page opens.
2. Click the checkbox corresponding to each capability that you want to include in the schedule.
3. Enter the start date, time zone, and the interval at which you want to run the job.
4. You can manage additional schedules using the following options:
   - To create a new schedule, click the **Add** button.
   - To delete a schedule, click the **Delete** button.
   - To enable or disable a schedule, click the **Enable Schedule** toggle button.

   Note: You can create a maximum of one schedule per capability that you enable. If you purged a catalog source or did not run the metadata extraction job, the catalog source job runs metadata extraction before running other scheduled capabilities.

   Note: To create a schedule, you need permissions on the connection to the source system. If you lose permissions on the connection after you create a schedule, the scheduled jobs continue to run.
5. Click **Save** to save the schedule.

Monitor job status

After the job runs, you can monitor the status of the job on the **Overview** page of the job.

For more information about job monitoring, see *Administration*.

Step 5. Assign reference catalog source connections to endpoint catalog source objects

When you run the catalog source job, if the catalog source references another source system, a reference catalog source and connection get created that point to the reference source system. To view the complete

lineage for your catalog source, you can perform connection assignment from the reference catalog source connection to the objects in the reference source system. A reference source system might be a database, such as Snowflake. You must first create and run an endpoint catalog source that connects to the reference source system.

Before you assign a connection, ensure that you have created and run an endpoint catalog source for each reference source system.
Note: If the source schema contains case-sensitive tables or if the reference objects contain multiple objects with the same name in different cases, perform case-sensitive connection assignment to get correct lineage.

If you enabled the lineage discovery capability for your catalog source, you can either curate the CLAIRE recommended endpoint objects on the **Related Catalog Sources** tab or assign connections manually.

For more information about related catalog sources and lineage discovery, see *Lineage discovery* in the *Administration* help.

1. On the **Configure** page, select the **Lineage** tab, and then select the **Lineage Discovery** tab. On the **Catalog Sources** panel, select the required catalog source and click the **Assign Connections** tab.

   The **Assign Connections** tab displays a list of assigned and unassigned connections along with details for each connection. Use filters to view the connections based on the connection names. Click the **Add Filter** menu to add filters.

2. Select the connection to the reference source system and click **Assign**.

   The connection name appears prefixed to the reference catalog source name on the **Hierarchy** tab of your catalog source in Data Governance and Catalog.

   The **Assign Connection** dialog box appears with a list of recommended objects from the endpoint catalog sources. Click **All** to view all endpoint catalog source objects.

3. In the **Assign Connection** dialog box, select one or more catalog sources to assign to the selected connection and click **Assign**.

   You can filter the list in the **Assign Connection** dialog box by name, type, or endpoint.

   You can assign Oracle and Snowflake as referenced source systems. To create a connection assignment to Snowflake SQL Script catalog sources, the referenced catalog sources must belong to the Schema class type.

   When you click **Assign**, Metadata Command Center creates links between matching objects in the connected catalog sources, and it calculates the percentage of matched and unmatched objects. The higher the percentage of matched objects, the more accurate the lineage that you view in Data Governance and Catalog.

4. Run the catalog source job again. If you configured the catalog source job to run on a regular schedule, the next scheduled run picks up the updated details. If you didn't configure a schedule, run the catalog source job again to view complete lineage.

CHAPTER 4

View results in Data Governance and Catalog

After Metadata Command Center runs a job, you can view the results in Data Governance and Catalog where the catalog source and its elements are called technical assets. You can view a catalog source as a hierarchy. Expand each technical asset to see its components.

When referenced source systems are connected to a catalog source, you can expand the hierarchy to see details about the technical asset's component elements.

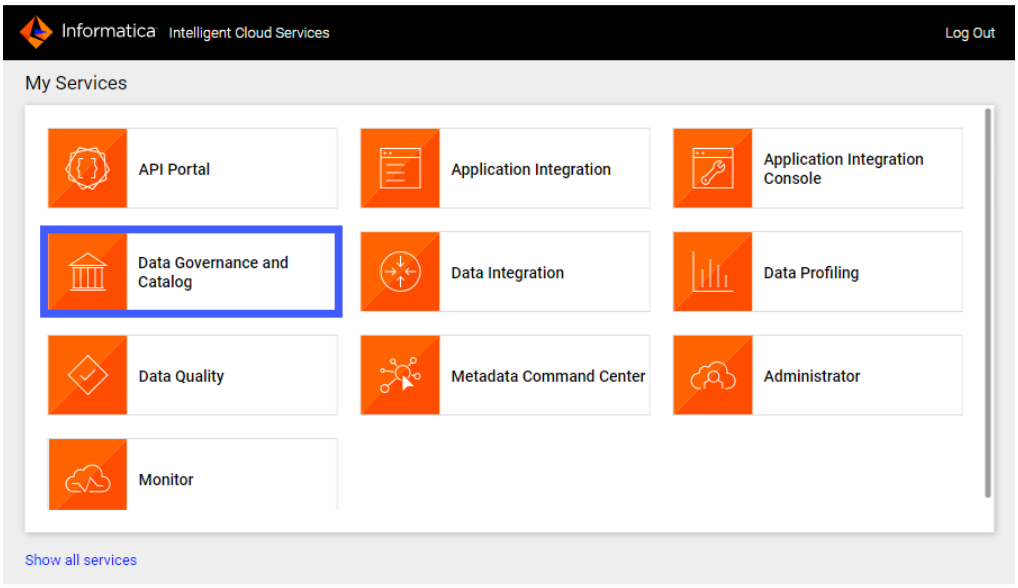
You can view the data lineage of an asset contained within a catalog source to see individual elements such as data sources, calculations, and filters. When you view data lineage, you can see the individual upstream elements that contribute data or expressions to each component of a data flow or catalog source.

View metadata extraction results

After a job runs in Metadata Command Center, view the results in Data Governance and Catalog. You can view details about source system contents as hierarchical displays and trace data lineage across task flows.

1. Log in to Informatica Intelligent Cloud Services and select Data Governance and Catalog from the **My Services** page.

   The following image shows the **My Services** page:



2. On the Data Governance and Catalog home page, click the number in the **Technical Assets** panel.

   The **Technical Assets** page opens.

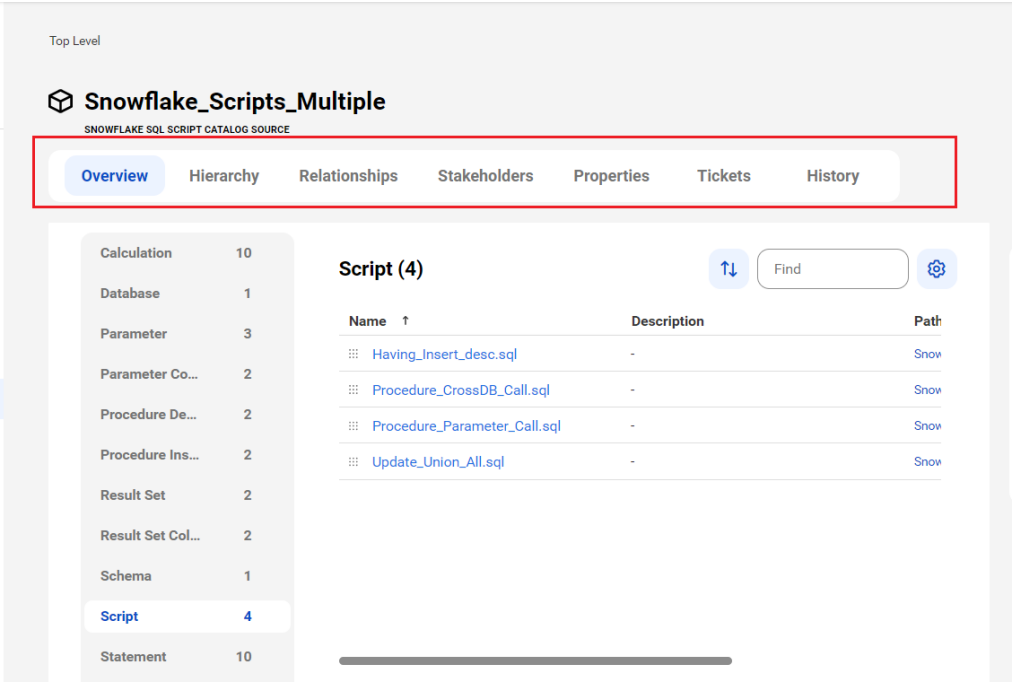
3. Select **Catalog Source** in the **Filter** list.

   The list of catalog sources opens.

4. Search for the catalog source from which you extracted metadata, and click the name.

   The **Overview** tab of the asset opens.

The following image shows a sample asset page:



5. View the asset from different perspectives by clicking on the tabs.

   Note: If you run catalog sources created prior to the July 2023 release with Retain as the **Metadata Change Option**, duplicate calculations appear in Data Governance and Catalog. To avoid duplicate calculations and to view complete lineage information, run catalog sources created prior to the July 2023 release with Delete as the **Metadata Change Option**.

For more information about working with assets, see "Working with Assets" in *Cloud Data Governance and Catalog* online help.

View data lineage

Data lineage is a visual representation of the flow of data across the systems in your organization. Lineage depicts how the data flows from the system of its origin to the system of its destination.

Data lineage views are available for technical assets in the catalog source. You can view lineage at the catalog source, data set, or data element level.

The lineage at the catalog source level shows how data flows from one catalog source to another. The lineage at the data set and the data element levels show how other technical assets such as files or tables contribute to the selected asset.

If linking catalog sources is available for your catalog source, you can use Metadata Command Center to generate data lineage based on rules or by generating automated lineage with CLAIRE. You can choose source and target catalog sources and objects to link and generate lineage.

To determine whether linking catalog sources is available for your catalog source, navigate to the **Configuration** tab of the **Link Catalog Sources** page. The catalog source must appear in the list of source and target catalog sources.

For information about linking catalog sources, see *Link catalog sources* in the Administration help.

View lineage at the catalog source level

The catalog source level shows how data flows from one catalog source to another with the lineage aggregating data from the data set and data element levels.

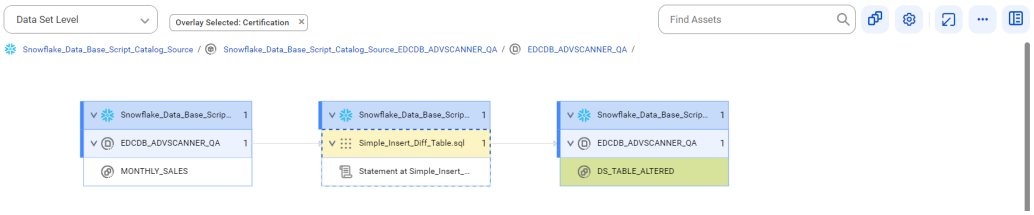
To view data lineage at the catalog source level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Catalog Source Level**.

View lineage at the data set level

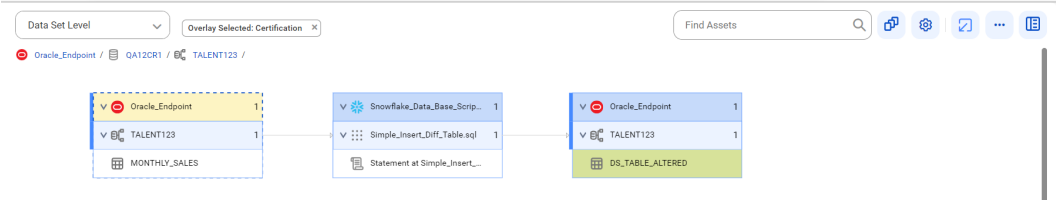
The data set level displays individual sets of data in the data flow.

To view data lineage at the catalog source level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Catalog Source Level**.

The following image shows table-level lineage where the DS_TABLE_ALTERED reference table gets data from the MONTHLY_SALES referenced table before connection assignment:



The following image shows table-level lineage where the DS_TABLE_ALTERED table gets data from the MONTHLY_SALES table after connection assignment:

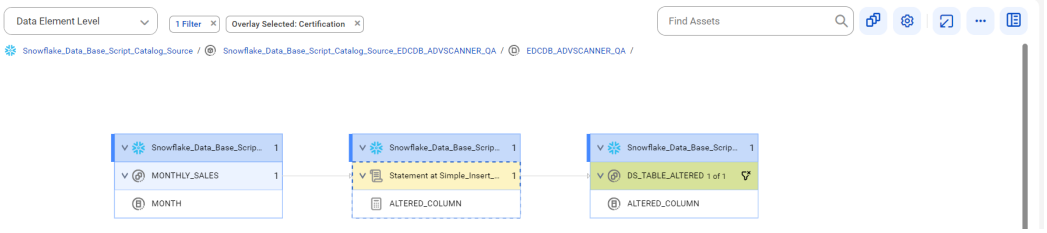


View lineage at the data element level

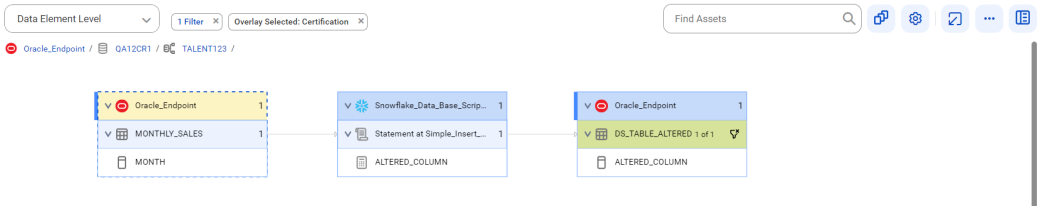
The data element level displays details of the data set level. At the data element level, you can see the input sources for expressions or commands and calculations or transformations on the data.

To view data lineage at the data element level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Data Element Level**.

The following image shows column-level lineage where the ALTERED_COLUMN reference column gets data from the MONTH referenced column before connection assignment:



The following image shows column-level lineage where the ALTERED_COLUMN column gets data from the MONTH column after connection assignment:



APPENDIX A

Managing catalog sources

You can update, copy, delete, purge, and run a catalog source. You can also grant configuration permissions for catalog sources to different roles in your organization.

On the **Catalog Sources** page of the **Explore** page, you can select a catalog source to see its overview and related catalog sources. The **Overview** tab displays the capabilities, filters, stakeholders, asset groups, and schedules enabled for a catalog source. The **Related Catalog Sources** tab displays a list of related catalog sources that are either assigned manually using connection assignment or are curated based on CLAIRE recommendations using the lineage discovery capability. You can view related catalog sources and accept or reject CLAIRE recommendations for catalog sources that are run with the lineage discovery capability.

Cloning a catalog source configuration

You can clone the configuration of a catalog source to create another catalog source with the same configuration.

Note: Cloning doesn't copy the content of the catalog source.

1. In Metadata Command Center, go to the **Explore** page.
2. Select **Catalog Sources** from the menu.
3. You can clone a catalog source in one of the following ways:
   - Hover your mouse over the catalog source that you want to clone and click **Clone** from the **Action** menu.
   - Open the catalog source that you want to clone and click **Clone**.

   Note: By default, the cloned catalog source has the same name suffixed with _Clone. You can change the name if needed.
4. If needed, modify the configuration of the catalog source by updating the parameters on each tab.

   Cloning doesn't copy the content of sensitive fields. If the catalog source contains sensitive fields such as Client Secret, enter the values.
5. Click **Save**.

Deleting or purging a catalog source

Delete or purge a catalog source based on your business requirements.

When you delete a catalog source, you delete all the configuration specific to that catalog source, along with all the metadata that was extracted and any enrichment made to the associated metadata. Purging a catalog source deletes all the metadata that was extracted, along with any enrichment made to the associated metadata.

Note: Before you delete or purge a catalog source, manually unassign the connections that are assigned to that catalog source.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source that you want to delete or purge.
4. Click the **Action** menu, and select **Delete** or **Purge**.

   Note: You can't delete or purge a catalog source if the catalog source job is in the running state.
5. To confirm delete or purge, click **OK**.

   The catalog source is triggered to delete or purge. You can monitor the status of the job on the **Monitor** page.

   Note: If the deletion job succeeds, the catalog source disappears from Metadata Command Center. If it fails, some associated objects might still be present in Data Governance and Catalog. Such catalog sources continue to appear in the list of catalog sources in Metadata Command Center, but are marked as **Deleted**. Delete the catalog sources to remove them along with any remaining objects present in Data Governance and Catalog.

Purging obsolete objects from a catalog source

You can purge obsolete objects from a catalog source.

When you configure or run a catalog source, you can choose to deprecate objects that are deleted from a source system. If you delete objects from the source system or make changes to the filter, the objects imported into the catalog before the change move to the Obsolete lifecycle in the catalog. You can purge obsolete objects from a catalog source. This permanently deletes all objects with the Obsolete lifecycle status, along with associated enrichments, from the catalog.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source from which you want to purge obsolete objects.
4. Click the **Action** menu, and select **Purge Obsolete Objects**.

   A warning message appears on the page.
5. Click **Yes** to confirm.

   A confirmation message appears on the page and the purge job starts. You can monitor the status of the job on the **Monitor** page. If the job succeeds, all obsolete objects present in the catalog source disappear from the catalog.

   Note: If the job fails, some obsolete objects might remain in the catalog. To remove the remaining obsolete objects, run the purge job again.

Curating CLAIRE recommendations

You can view related catalog sources and accept or reject CLAIRE recommendations for a catalog source.

Applicable to catalog sources that generate lineage.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source for which you want to curate CLAIRE recommendations.
4. Hover your mouse over the catalog source and click **View Lineage Discovery** from the **Action** menu.

   Note: The **View Lineage Discovery** option is available for catalog sources that are run with the lineage discovery capability.

   The **Configure** page appears with the catalog source filtered on the **Lineage Discovery** tab of the **Lineage** tab.
   The **Related Catalog Sources** tab displays a list of related catalog sources.
5. Hover your mouse over a related catalog source and click **Accept** or **Reject** from the **Action** menu.

   For more information about lineage discovery and curating CLAIRE recommendations, see *Lineage discovery* in the *Administration* help.

Managing configuration permissions for catalog sources

As a service administrator, you can grant configuration permissions for catalog sources to different roles in your organization, and add specific users and user groups to those roles so that they can perform operational activities on a catalog source.

To manage permissions for catalog source configuration, verify that at least one role has the catalog source configuration **Set Permissions** functionality enabled for Metadata Command Center in Administrator. In Administrator, create the required roles, assign users and user groups to the roles, and specify appropriate permissions to the roles.

You can select roles and add specific users or user groups to those roles to grant the following levels of permission for catalog source configuration:

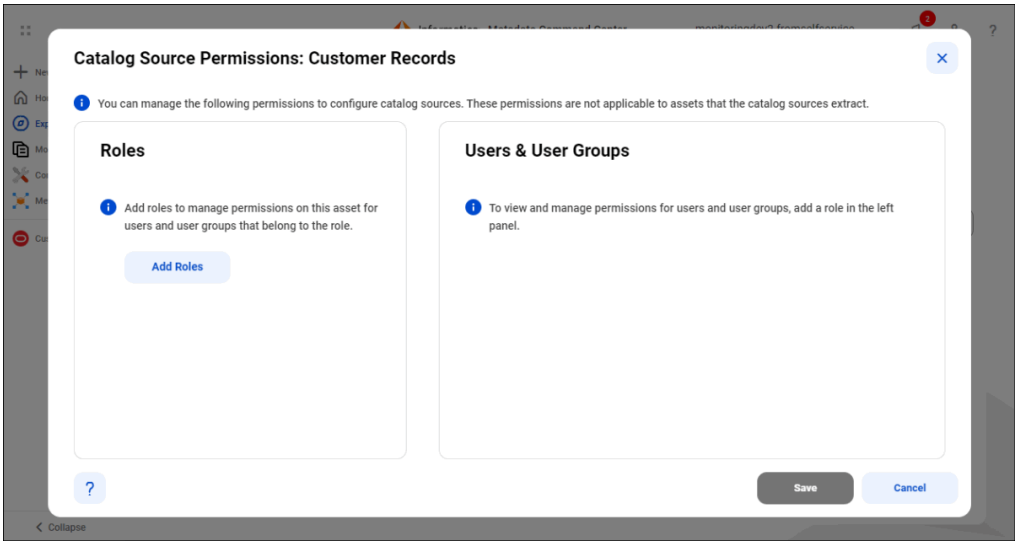
- Create
- Read
- Update
- Run
- Set Permissions

Based on the permissions granted to the roles, the selected users and user groups belonging to that role can perform appropriate operations for configuring catalog sources.

1. Go to the **Explore** page in Metadata Command Center.
2. From the list of catalog sources, search for the catalog source for which you want to manage permissions.

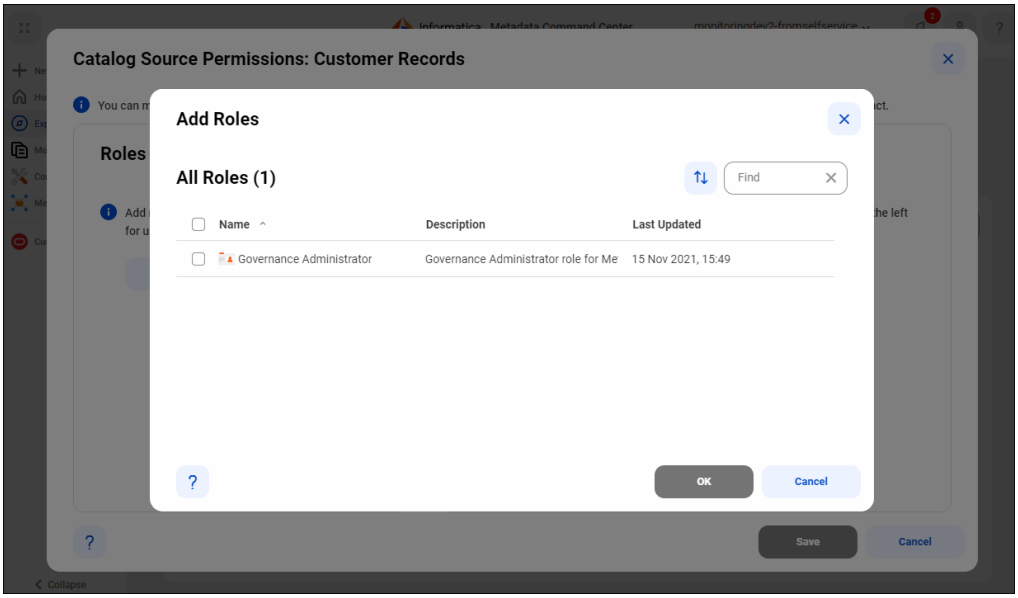
3. Click the **Action** menu, and select **Permissions**.

   The **Catalog Source Permissions** dialog box appears.



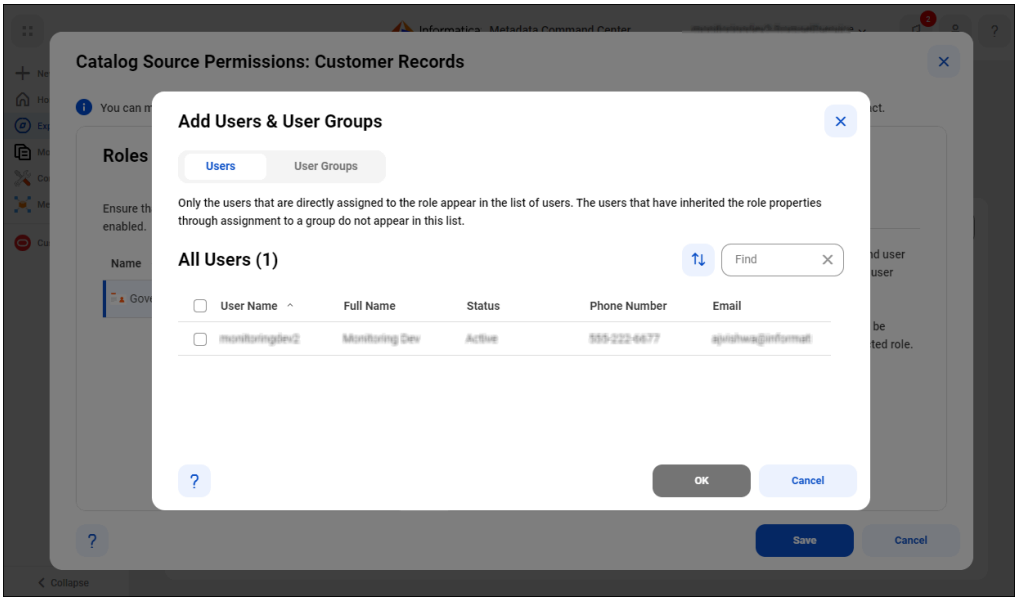
4. In the **Roles** panel, click **Add Roles**.

   The **Add Roles** dialog box appears.



5. Select one or more roles, and click **OK** to add the roles.
6. In the **Add Users & User Groups** panel, by default, all users and user groups that belong to the selected role are granted the role-defined permissions for the catalog source configuration.

To assign permissions only to the selected users and user groups that belong to this role, choose **Specific Users and User Groups**.



7. Select **Add Specific Users and User Groups**.

   The **Add Users & User Groups** dialog box displays the list of users and user groups that the administrator has assigned to the selected role.

8. Select one or more users or user groups to assign to the selected role, and click **OK**.

   Only the selected users and user groups belonging to the specified role will be granted the role-defined permissions for catalog source configuration.

9. View the assigned permissions for each user and or group, and click **Save**.